

CSE231 Operating System (Section A), Quiz 04 Monsoon 2025
Time allocated: 4pm – 4:20pm

Name	
Roll Number	

Instructions:

- This is a closed book and closed notes quiz. Please be aware of strict plagiarism policy.
- For questions requiring justification, please be as concise as possible. 2-3 sentences would be the ideal size of a justification. No extra pages will be provided.
- **NO EXTRA ANSWER SHEET WILL BE PROVIDED. USE THE GIVEN SPACE WISELY**

Question-1: Name the six segment registers available on x86 processor. Which ones of these are mandatorily used by any process on x86 hardware for a) operating on “.text” section, and b) maintaining the caller-callee relationship in method call chain. Assume that the processor **only** relies on segmentation-based approach for memory management. No justifications needed **[2 marks]**.

CS, SS, DS, ES, FS, and GS.

a) CS

b) SS

Marking scheme: $0.25 \times 8 = 2$ marks

Question-2: Which two components, related to memory translation, are automatically stored in the invisible part of a segment register on an x86 processor to define the segment's location and size **[1 mark]**. No justifications needed.

Base and Bound. Marking scheme: $0.5 \times 2 = 1$ mark

Question-3: For an 8-bit OS on a system with 8 KB of DRAM and an 8-byte page size, calculate the total number of virtual and physical pages. Also, determine how many bits are required to uniquely identify the Virtual Page Number (VPN) and the offset in any virtual address. **[3 marks]**.

$VM = 2^8$

Page size = 2^3

$VPNs = 2^8 / 2^3 = 2^5$ // 1 mark

$PM = 2^{13}$

$PPNs = 2^{13} / 2^3 = 2^{10}$ // 1 mark

Since there 2^5 VPNs, bits needed to uniquely identify VP = 5, // 0.5 mark

Bits for offset = 3. // 0.5 mark

Question-4: For the system configuration presented in Question-3, assume there is a TLB that can hold up to four entries. How many TLB misses and hits would occur while iterating through an integer array with eight elements? Assume no other processes are running **[2 marks]**. Briefly justify.

Two consecutive elements in an array occupies 1 page (4+4 bytes) // 0.5 mark

Total pages needed = 4. Each page will have one miss and one hit // 0.5 mark

TLB miss = 4. // 0.5 mark

TLB hits = 4. // 0.5 mark

Question-5: Why an inverted page table named as “inverted”? **[2 marks]**

It is called inverted as total entries is equal to the total number of PPNs [No partial marking]
